

Violence Exposure Moderates Stress-Elicited Neurobehavioral Function in Young People

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Violence exposure has deleterious effects on emotional well-being, including higher rates of future mental illness. Adolescence is an important period of neural development within brain regions (e.g., prefrontal cortex) that support emotional processes. The relationship between brain activity and emotion may vary with violence exposure. Thus, this study investigated the relationship between violence exposure, stress-elicited brain activity, and emotion in young people. Violence exposure was measured four times from 11 to 19 years of age. Participants ($n = 301$) returned 1 year later (age = 20) to complete mental health (i.e., anxiety, depression, and posttraumatic stress) questionnaires and the Montreal Imaging Stress Task during behavioral (e.g., skin conductance response and cortisol) and neuroimaging data collection. Data were collected from 2004 to 2018. Violence exposure was positively associated with mental health symptoms. Further, violence exposure moderated the relationship between stress-elicited dorsolateral prefrontal cortex activity and depression, cortisol, and skin conductance response. These findings suggest that violence exposure moderates the relationship between stress-elicited brain function and emotion-related behavior in young people. These findings provide novel insight into neural processes that may underlie the relationship between prior violence exposure and emotional function, which may have important implications for mental health.

Keywords: functional MRI, violence exposure, stress reactivity, depression, prefrontal cortex

Adolescence is an important period in a child's development and can have a lasting impact on emotional function that may persist well into adulthood (Buckholdt et al., 2015; Cook et al., 2018; Heinze et al., 2017; Lo Iacono & Carola, 2018; Pattwell & Bath, 2017). For example, adolescent violence exposure has harmful effects on the emotional well-being of children, increasing their risk for internalizing problems (e.g., anxiety, depression, and posttraumatic stress) later in life (Buckholdt et al., 2015; Cook et al., 2018; Heleniak et al., 2018; Källström et al., 2020; Marusak et al., 2015; Mrug & Windle, 2010).

Specifically, adolescent violence exposure appears to increase risk for internalizing disorders by disrupting emotion processes (e.g., emotion regulation) that do not fully develop until young adulthood (Buckholdt et al., 2015; Lo Iacono & Carola, 2018; Marusak et al., 2015). Given that emotion expression and regulation are supported by a neural circuit that includes the dorsolateral, dorsomedial, ventrolateral, and ventromedial prefrontal cortex (PFC), inferior parietal lobule (IPL), hippocampus, and amygdala (Casey et al., 2019; Cheng et al., 2003; Goodman, Harnett, Wheelock, et al., 2018; Harnett et al., 2019;

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Orem et al., 2019; Pattwell & Bath, 2017; Teffer & Semendeferi, 2012; Wang et al., 2005; Wheelock et al., 2016, 2018), the relationship between these brain regions and emotional function may be moderated by the amount of violence exposure.

Emotion processes are supported by a neural network that includes the dorsolateral PFC, dorsomedial PFC, ventrolateral PFC, ventromedial PFC, IPL, hippocampus, and amygdala (Casey et al., 2019; Goodman, Harnett, Wheelock, et al., 2018; Harnett et al., 2019; Orem et al., 2019; Radley et al., 2011; Silvers & Guassi Moreira, 2019; Wheelock et al., 2016, 2018). Specifically, the hippocampus processes contextual information about the environment to support emotional function (Casey et al., 2019; Goodman et al., 2016; Pattwell & Bath, 2017; Purcell et al., 2021). The hippocampus projects to the dorsolateral PFC, a region that supports emotional learning and memory (Åhs et al., 2015; Goodman, Harnett, & Knight, 2018; Purcell et al., 2021). The learning-related anticipatory processes supported by the dorsolateral PFC then modulate threat and stress-related function via projections to the IPL and dorsomedial PFC (Åhs et al., 2015; Casey et al., 2019; Choi et al., 2017; Dark et al., 2023; Harnett et al., 2018; Silvers & Guassi Moreira, 2019; Wheelock et al., 2016). The dorsomedial PFC interacts with the ventromedial and ventrolateral PFC, and together they regulate amygdala activity (Casey et al., 2019; Delgado et al., 2008; Goodman, Harnett, & Knight, 2018; Goodman, Harnett, Wheelock, et al., 2018; Schultz et al., 2012; Silvers & Guassi Moreira, 2019; Wheelock et al., 2016, 2018). The amygdala supports processes important for emotion expression, and it projects to other subcortical brain regions (e.g., hypothalamus) to control the peripheral expression of emotion (Casey et al., 2019; Cheng et al., 2006, 2007; Dark et al., 2023; Fendt & Fanselow, 1999; Goodman, Harnett, & Knight, 2018; Knight et al., 2005; LeDoux et al., 1990; Orem et al., 2019). Together, these brain regions form a neural circuit that supports emotional processes that appear to underlie susceptibility and resilience to internalizing disorders (e.g., anxiety, depression, and posttraumatic stress).

Adolescence is an important stage of maturation during which chronic stressors may alter neural and emotional development (Casey et al., 2019; Cook et al., 2018; Heleniak et al., 2018; Marusak et al., 2015). For example, chronic stressors (e.g., violence exposure) modify emotional expression and regulation processes that affect the ability to manage stress (Cook et al., 2018; Heleniak et al., 2018; Marusak et al., 2015). Further, prior research suggests that adolescent violence exposure promotes the development of coping strategies that have short-term benefits, but long-term consequences for adult mental health (Buckholdt et al., 2015; Danese & McEwen, 2012). For example, violence exposure is associated with avoidance, isolation, negative self-evaluations, and aggression toward others (Kliewer & Sullivan, 2008; Seiffge-Krenke, 2000). These strategies may serve as a buffer, protecting the adolescent from physical or psychological risks (e.g., violence exposure; DeCarlo Santiago & Wadsworth, 2009; Kliewer & Sullivan, 2008; Reid-Quinones et al., 2011). Although these coping mechanisms may provide short-term emotional relief (DeCarlo Santiago & Wadsworth, 2009; Reid-Quinones et al., 2011), they also appear to increase symptoms of internalizing disorders (i.e., depression, anxiety, and posttraumatic stress) over time (Kliewer & Sullivan, 2008; Seiffge-Krenke, 2000). Adolescence is also an important period of neural development, including the development of brain regions (e.g., PFC, amygdala, hypothalamic–pituitary–adrenal axis) that support the expression and regulation of the peripheral emotional response (Danese & McEwen, 2012; Lo Iacono & Carola, 2018; Teffer &

Semendeferi, 2012). Thus, adolescence may be a period of particular vulnerability to chronic stressors (e.g., violence exposure) due to the brain's ongoing development. In fact, adolescent violence exposure is associated with heightened baseline and stress-induced hypothalamic–pituitary–adrenal axis activity, which may also be linked to emotional dysfunction as adolescents emerge into adulthood (Danese & McEwen, 2012; Radley et al., 2011). Therefore, the relationship between neural function and subsequent stress-elicited behavior may be moderated by violence exposure.

Brain regions that include the PFC, IPL, hippocampus, and amygdala undergo extensive maturation during adolescence, which appears to impact emotional function (Danese & McEwen, 2012; Lo Iacono & Carola, 2018). These brain regions form a frontoparietal–hippocampal–amygdala network that supports important emotion-related processes (e.g., emotion expression and regulation), which continue to develop well into young adulthood (Casey et al., 2019; Cheng et al., 2006; Danese & McEwen, 2012; DeSerisy et al., 2021; Goodman, Harnett, Wheelock, et al., 2018; Harnett et al., 2018; Knight et al., 2005; Kolk & Rakic, 2022; Wheelock et al., 2018). For example, emotion regulation abilities appear to develop in parallel with the maturation of the PFC during adolescence (Caballero et al., 2016; Kolk & Rakic, 2022; Kundu et al., 2018; Teffer & Semendeferi, 2012). Further, developmental changes during adolescence in the functional connectivity of the PFC and limbic system (i.e., hippocampus, amygdala, and hypothalamus) are also associated with emotional function (Casey et al., 2019).

Violence exposure may alter brain function by modifying neural processes that support healthy emotional function. For example, stress triggers neurobehavioral processes that are normally adaptive, but become disruptive when stress exposure is prolonged (Danese & McEwen, 2012; Lo Iacono & Carola, 2018; Radley et al., 2011). Specifically, chronic stress during adolescence may promote maladaptive neural (e.g., weak PFC–amygdala connectivity) and psychophysiological (e.g., high baseline cortisol levels) function (Casey et al., 2019; Danese & McEwen, 2012; Radley et al., 2011). Thus, the chronic stress associated with high levels of violence exposure may alter the relationship between neural processes and the emotional response to stress. Taken together, this prior work suggests that emotional and neural maturation during adolescence may make this developmental period particularly vulnerable to chronic stressors (e.g., violence exposure).

Adolescent stress exposure has been linked to negative mental health outcomes in adulthood (Buckholdt et al., 2015; Cook et al., 2018; Mrug & Windle, 2010; Pattwell & Bath, 2017). For example, a single exposure to a significant stressor (e.g., violence exposure) in adolescence increases the risk for developing an internalizing disorder (e.g., depression, anxiety, and posttraumatic stress) by 30% (Pattwell & Bath, 2017). Further, there appears to be a dose–response relationship between adolescent violence exposure and adult psychopathology. Specifically, the greater the exposure to violence during adolescence, the greater the risk of adult psychopathology (Felitti et al., 1998). Emotion-related neural circuitry (e.g., frontoparietal–hippocampal–amygdala network) appears to underlie susceptibility to psychopathology (Åhs et al., 2015; Davidson, 2000; Fitzgerald et al., 2006; Kumari et al., 2003). Further, the relationship between the function of this brain circuitry and psychopathology may vary with the amount of violence exposure. For example, chronic activation of the hypothalamic–pituitary–adrenal axis in adolescence (e.g., linked to greater adolescent violence exposure) is associated with greater stress susceptibility in adulthood (Danese & McEwen, 2012;

Radley et al., 2011). Furthermore, adolescent adverse life events have been linked to lower frontoparietal network activity and connectivity (Dark et al., 2020; Harnett et al., 2019; Lo Iacono & Carola, 2018; Teicher & Samson, 2016; Wheelock et al., 2016). Thus, the link between frontoparietal function and emotion regulation deficits that develop as adolescents emerge into adulthood may be moderated by violence exposure (Dark et al., 2020; Harnett et al., 2018). Further, emotion regulation deficits associated with violence exposure may increase susceptibility to psychiatric disorders. In fact, psychiatric symptoms (e.g., anxiety, depression, and posttraumatic stress) are greater in those who have experienced violence (Danese & McEwen, 2012; Harnett et al., 2018; Teicher & Samson, 2016). Thus, greater stress susceptibility may lead to greater vulnerability to internalizing disorders, such as anxiety, depression, and posttraumatic stress disorder (PTSD; Danese & McEwen, 2012; Radley et al., 2011; Teicher & Samson, 2016). Therefore, prior violence exposure may moderate the risk of adult psychopathology.

The present study investigated the relationship between prior violence exposure and the neural response to stress. Specifically, we investigated the relationship violence exposure (assessed between 11 and 19 years of age) has with neural reactivity to stress and emotional function (e.g., anxiety, depression, posttraumatic stress, and psychophysiology) in emerging adulthood (20 years of age). We expected prior violence exposure to moderate the relationship between stress-elicited activity within the frontoparietal–hippocampal–amygdala network and emotional function. Taken together, the present study provides novel insight into the interrelationships between violence exposure, brain reactivity to stress, and emotional function in young people.

Method

Participants

Participants were recruited from the Healthy Passages study, a longitudinal study ($n = 1,594$) of risk factors and outcomes of early life experiences (Mrug & Windle, 2009; Windle et al., 2004). Participants were initially recruited from 5th-grade classrooms in local public schools and interviewed at three time points (Wave 1: age = 11.17 ± 0.46 ; Wave 2: age = 13.03 ± 0.46 ; and Wave 3: age = 16.20 ± 0.47 ; $M \pm SD$). Wave 1 data were collected between 2004 and 2006, Wave 2 data were collected between 2006 and 2008, and Wave 3 data were collected between 2009 and 2011. An additional interview (Wave 4: age = 19.59 ± 1.13 years) was conducted between 2012 and 2017 as a part of the present study. Neuroimaging was completed shortly after the Wave 4 interview on ~22% ($n = 350$) of the initial Healthy Passages cohort. Neuroimaging data were collected between 2013 and 2018. Sample size was determined based on an a priori power analysis for a larger project, within which the present study was completed. Specifically, the test of not-close fit (MacCallum et al., 1996), with an α of .05 and an estimated sample size of 240, yielded a power to detect well-fitting models of .82. The ratio between the estimated sample size and number of free parameters was 8:1, using 28 free parameters of the model for the overarching study. This ratio was above the recommended 5:1 to obtain stable parameter estimates (Bentler & Chou, 1987). Thus, the power to detect significant small-to-medium effects was above 0.8 (Fritz & Mackinnon, 2007). Given that only a portion of the initial cohort completed neuroimaging, we assessed demographic differences between participant groups who did and did not complete the

neuroimaging session. There was no difference in the proportion of male and female participants who did versus did not complete neuroimaging, $\chi^2(1) = 0.11$, $p = .75$. A greater proportion of Black than White participants completed neuroimaging, $\chi^2(1) = 7.85$, $p < .01$. Violence exposure (see below for more information) at Wave 1 did not differ between the participants that did ($M = 0.90 \pm 0.98$) versus did not ($M = 1.01 \pm 1.00$) complete neuroimaging, $t(1483) = 1.73$, $p = .08$.

Additional participant characteristics were collected and can be found in Table 1. Specifically, family income was collected across Waves 1–3. Family income was computed as a percentage of the federal poverty line and averaged across Waves 1–3 as an index of socioeconomic status across adolescence. Substance use (i.e., alcohol, tobacco, and cannabis use across Waves 1–4), medical conditions (Wave 4), and psychotropic medication usage (Wave 4) have been included in Table 1.

Data from 49 participants were excluded from analyses due to motion, artifacts, or incomplete data. The final sample consisted of 301 participants (149 male, 152 female; 201 Black, 100 White; age = 20.03 ± 1.51 years, age range 17–23 years; Table 1). Exclusion criteria for the present study included blood disorders (e.g., anemia), pregnancy, left-handedness (i.e., due to brain laterality differences), a history of psychosis, seizures, or head injury, and contraindications for magnetic resonance imaging (MRI). Outliers were truncated to 3SD from the mean for all variables of interest. All participants provided written informed consent, and all procedures were approved by the University of Alabama at Birmingham Institutional Review Board.

Table 1
Participant Demographics

Demographic variable	Category	<i>N</i> (%)	<i>p</i>
Sex	Male	149 (49.5)	.863
	Female	152 (50.5)	
Race	Black	201 (66.8)	<.001
	White	100 (33.2)	

Demographic variable	<i>N</i>	<i>M</i>	<i>SD</i>	Range
Scan age	301	20.03	1.51	17–23
Differential stress ratings	288	10.58	8.14	−6.76–31.00
Violence exposure	301	0.85	0.67	0.00–3.03
SES	287	272.53	255.83	0.00–1384.08

Demographic variable	<i>N</i> (yes)	<i>N</i> (no)
Psychotropic medication	17 (5.6%)	284 (94.4%)
Medical condition	37 (12.3%)	264 (87.7%)
Previous alcohol use	238 (79.9%)	60 (20.1%)
Previous tobacco use	140 (47%)	158 (53%)
Previous cannabis use	144 (48.3%)	154 (51.7%)

Note. Values for sex, race, psychotropic medication status, and medical condition reflect the number of participants (and %) within the neuroimaging sample ($N = 301$). *p* values for sex and race represent the difference in the proportion of each group (e.g., male vs. female participants). Values for sample size (*N*), mean, and standard deviation (*SD*) are presented for scan age (years), differential stress ratings (stress–control MIST), violence exposure, and socioeconomic status. Substance use data were not collected for three participants. Therefore, substance use data for these participants are not included in this table. SES = socioeconomic status; MIST = Montreal Imaging Stress Task.

Montreal Imaging Stress Task

Participants completed a functional MRI (fMRI) study of psychosocial stress using a modified version of the Montreal Imaging Stress Task (MIST), a psychosocial stress task designed for the functional neuroimaging environment (Dedovic et al., 2005; Wheelock et al., 2016). The study used an event-related design comprised of two separate conditions presented over two fMRI scans (control MIST and stress MIST). Each scan included 54 trials and was 7 min and 54 s in duration. Each trial was separated by a 1- to 3-s intertrial interval. During each trial (6 s duration), participants answered unique math problems with choices that ranged from 0 to 9, presented at the bottom of the visual display. Answer choices were presented for 0.5–5 s and were followed by a fixation cross (0.5–5 s duration). The trial ended with 0.5 s of visual feedback (“right,” “wrong,” or “timeout”).

Participants were familiarized with the math task prior to the scanning session by completing a set of practice problems outside of the scanning environment. Performance on the practice trials was used to establish the difficulty level of the problems that were presented during scanning. Participants answered math problems that ranged in difficulty from easy (addition or subtraction of two single-digit numbers), medium easy (addition or subtraction of three single-digit numbers), medium hard (addition or subtraction of four single- or double-digit numbers), and hard (addition, subtraction, and multiplication of four single- or double-digit numbers) during practice. No participants were able to successfully complete the medium hard or hard difficulty levels within the time limit for the task (i.e., 5 s) during practice. Therefore, all participants received either the easy or medium easy version of the task during fMRI. The difficulty level of the math problems remained constant across control and stress conditions of the task.

Investigators attempted to lower participants’ stress level prior to the control condition by telling them that “It is okay if you do not answer all of the problems correctly.” In addition, participants received positive auditory feedback at four separate time points during the control condition of the task in an effort to lower stress levels. This feedback consisted of phrases like “Good job. You’re doing just fine. Keep up the good work.” In contrast, investigators tried to raise stress levels prior to the stress condition of the MIST. Participants were told that others had answered more than 80% of the questions correctly and that they must do the same for their data to be used in the present study. In addition, participants received negative auditory feedback at four time points during the stress condition of the task, which consisted of phrases such as “You are not doing as well as we had hoped. Please do your best to answer these correctly.” Failure was ensured by controlling the time in which the participant could respond using a stair-step procedure. The stair-step procedure was implemented by decreasing the available time by 0.5 s for each correct answer and increasing the time by 0.5 s for each incorrect answer. Thus, the time available to answer math problems could vary between 0.5 s (minimum) and 5.0 s (maximum) in 0.5 s increments. Consequently, participants were able to answer approximately 50% of the problems correctly. Additional details of this version of the MIST have been published previously (Goodman et al., 2016; Orem et al., 2019; Wheelock et al., 2016).

Stress Rating

A measure of self-reported stress level was used as an index of each participant’s emotional response to the control and stress

conditions of the MIST (Wheelock et al., 2016). Following the MIST, participants completed a questionnaire consisting of eight statements. Four of the statements were positively worded such as “I felt calm,” and four were worded negatively such as “I felt stressed.” Participants rated the applicability of each statement on a 5-point scale (1 = *not at all*, 3 = *moderately*, 5 = *extremely*). Responses were then summed as an index of stress that ranged from eight to 40.

Heart Rate

Heart rate was monitored during the scanning session to assess the emotional response to control and stress conditions of the MIST. Cardiac data were collected using an MRI compatible photo-plethysmograph (Siemens; Munich, Germany) placed on the distal phalanx of the index finger of the left hand. Cardiac time series data were recorded at 50 Hz using a Siemens Physiological Monitoring Unit. QRSTool was used to identify the peak pulse waveforms within each cardiac time series, and CMetX (Allen et al., 2007) was used to calculate the heart rate for control and stress conditions of the MIST.

Skin Conductance

Skin conductance response (SCR) was monitored during the scanning session as a measure of the peripheral emotional response to the stress and control conditions. Skin conductance data were collected using an MRI compatible physiological monitoring system (Biopac Systems; Goleta, CA, United States) as described in prior work (Knight & Wood, 2011; Orem et al., 2019). Skin conductance was sampled at 10 kHz using a pair of disposable, 1 mm diameter, radio translucent electrodes (Biopac Systems; Goleta, CA, United States). Isotonic recording electrode gel (Gel101, Biopac Systems; Goleta, CA, United States) was applied to the electrodes, which were then placed on the thenar and hypothenar eminence of the left hand. Skin conductance data were low pass filtered at 1 Hz and down sampled to 250 Hz using Acqknowledge 4.1.0 software. The down sampled data were analyzed using SCRalyze toolbox (Version b2.1.8) and bandpass filtered with a first-order Butterworth filter (highpass cutoff 0.0159 Hz, lowpass filter of 1.0 Hz; Bach et al., 2009). Skin conductance data were then further down sampled to a 10 Hz sampling rate and z-transformed and mean centered. SCRs to math events were determined using a general linear model with an SCR function without a time or dispersion derivative (Bach et al., 2009).

Cortisol

Two saliva samples were collected to assess the cortisol response to the MIST. The first sample (baseline) was collected 1 hr after the participant arrived, just prior to the scanning session. The second sample (post-MIST) was collected 20 min after completion of the MIST. Saliva samples were collected using passive drool through a short straw into 2.0 ml cryovials. Samples were frozen at -80°C and shipped overnight to the Institute for Interdisciplinary Salivary Bioscience Research, where they were assayed for cortisol using a commercially available immunoassay (Salimetrics, LLC, Carlsbad, CA), using the manufacturer’s recommended protocol. The salivary sample volume was 25 μl with an assay range of sensitivity from 0.007 to 3 $\mu\text{g/dL}$. Intra and interassay coefficients of variation were, on average, less than 5% and 15%. All samples were tested in

duplicate, and the average of the duplicates was used in statistical analyses. Prior studies have sampled cortisol only during the afternoon to control for diurnal rhythms. However, we were unable to control for diurnal rhythms in this manner due to limitations in participant and MRI scanner availability. Therefore, data were collected between the hours of 8 a.m. and 5 p.m.

Violence Exposure

Each participant's violence exposure was assessed at Waves 1–4 (i.e., ages 11–19 years) using the Healthy Passages Violence Exposure measure (Dark et al., 2020). Participants were asked how frequently they had witnessed violence such as (a) threat of violence, (b) physical violence, and (c) threat or physical violence involving a weapon within the past 12 months. In addition, participants were asked how frequently they had been victimized by violence including (a) threat of violence, (b) physical violence, (c) threat or violence involving a weapon, and (d) violent injury requiring medical treatment within the past 12 months. Participants rated items on a 4-point frequency scale ranging from 0 to 3 (0 = *never*, 1 = *one time*, 2 = *a few times*, 3 = *many times*). Responses to witnessing and victimization questions were averaged separately at each wave, resulting in a “witnessing” score and a “victimization” score for each wave of the study. The witnessing and victimization scores within each wave were summed, providing a cumulative violence exposure score from 0 to 6 for each of the four waves of the study. The cumulative violence exposure scores from each wave of the study were then averaged together to produce the violence exposure variable that was utilized in the present study. The collection of violence exposure data, during Waves 1–4, preceded the collection of all neural, physiological, and stress rating data, which were collected shortly after Wave 4.

Anxiety

The Physiological Anxiety subscale of the Revised Children's Manifest Anxiety Scale (RCMAS; Lee & Piersel, 1987) was used to assess anxiety during Waves 1–4. Anxiety data from Wave 4 were used in subsequent analyses as an index of anxiety in adolescents emerging into adulthood. The questionnaire included 10 questions such as “Do you often have trouble making up your mind?” and “Do you often have bad dreams?” Participants answered each question as either “No” or “Yes” (scored as *No* = 0 and *Yes* = 1). Responses were then summed to create an index of anxiety that ranged from 0 to 10. The RCMAS was used across all four waves to maintain continuity across time. Although the RCMAS has not been validated in adult samples, RCMAS questions largely overlap with those of the General Anxiety Disorder–7 Scale (Spitzer et al., 2006) and Hamilton Anxiety Rating Scale (Thompson, 2015), which have been validated in adults.

Depression

Symptoms of depression were assessed during Waves 1–4 using a modified version of the Major Depression Subscale of the Diagnostic Interview Schedule for Children (DISC; Shaffer et al., 2000). Depression data from Wave 4 were used in the present study as an index of depression in adolescents emerging into adulthood. Questions included “During the past 12 months, has there been a time when nothing was fun for you and you just weren't interested

in anything?” and “During the past 12 months, has there been a time when you felt you couldn't do anything well?” The DISC includes seven questions, but one question about self-harm was removed due to concerns raised by the Institutional Review Board. Thus, participants answered six questions either “No” or “Yes” (scored as *No* = 0 and *Yes* = 1). Answers were then summed to create an index of depressive symptoms ranging from 0 to 6. The DISC was used across all four waves to maintain continuity across time. Although the DISC has not been validated in adult samples, questions from the DISC largely align with those of the Hamilton Depression Rating Scale (Sharp, 2015) and Beck Depression Inventory (Beck et al., 1961), which have been validated in adults.

Posttraumatic Stress

Posttraumatic stress symptoms were assessed at Wave 4 using the Childhood PTSD Symptom Scale (Foa et al., 2001). The questionnaire included 17 questions related to symptoms such as “flashbacks or nightmares.” Participants answered each question on a 0 to 3 scale (0 = *not at all*, 1 = *some of the time*, 2 = *most of the time*, 3 = *almost always*). Responses were averaged across the 17 questions to produce a total score, ranging from 0 to 3, for each participant. The Childhood PTSD Symptom Scale has not been validated in adult samples. However, the Childhood PTSD Symptom Scale questions are similar to those of the Clinician Administered PTSD Symptom Scale for the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition (Weathers et al., 2018), which has been validated in adults.

Functional MRI

Functional MRI data were collected on 3T Siemens Allegra ($n = 251$) and Prisma ($n = 50$) scanners. High resolution anatomical images (magnetization prepared rapid gradient echo) were acquired as an anatomical reference, Allegra: (T1 weighted; repetition time = 2,300 ms; echo time = 3.9 ms; field of view = 25.6 cm; matrix = 256 × 256; slice thickness = 1 mm; 0.5 mm gap) and Prisma: (T1 weighted; repetition time = 2,300 ms; echo time = 2.98 ms; field of view = 25.6 cm; matrix 256 × 256; slice thickness = 1 mm; 0.5 mm gap). Blood oxygenation-level-dependent fMRI data were acquired using a gradient-echo echoplanar pulse sequence (repetition time = 2000 ms; echo time = 30 ms; field of view = 24 cm; matrix 64 × 64; slice thickness = 4 mm) during two scans of stimulus presentations. All fMRI data were preprocessed using the Analysis of Functional NeuroImages (AFNI) software package (Cox, 1996). Spatial smoothing was completed using a 4-mm full-width-at-half-maximum Gaussian kernel. Functional MRI data were slice-time corrected and registered to the fifth volume of the first scan. First-level analyses modeled joystick movement, button presses, auditory feedback, visual feedback, and used the six head motion parameters as regressors of no interest. Math problems were used as the regressor of interest. Visual feedback events were modeled using an instantaneous response function, while the hemodynamic response functions for math trials and auditory feedback events were modulated using the duration of the response time (trial onset to answer selection or timeout) and the duration of the auditory feedback recording, respectively. Additionally, motion correction was completed by excluding high motion images (i.e., volumes in which greater than 3% of voxels were considered outliers using AFNI's 3dToutcount). Participants with less than 80% usable volumes were excluded from data analysis ($n = 1$). Functional MRI data from math

events were resampled to 1 mm³ and normalized to the Montreal Neurologic Institute 152 template (Fonov et al., 2011).

Statistical Analyses

Statistical Package for the Social Sciences software was used to complete analyses of behavioral data. Specifically, paired *t*-tests were used to assess differences between the stress and control conditions of the MIST (Table 2). Additionally, Pearson's correlation was used to assess the relationship between violence exposure and behavioral measures (Table 3). Further, moderation analyses were conducted, using the Mediation and Moderation for Repeated Measures macro in Statistical Package for the Social Sciences (Montoya & Hayes, 2017), to assess whether the relationship between MIST condition (i.e., stress vs. control MIST) and stress-related behaviors were moderated by the amount of violence exposure. Significant moderation effects were then followed by simple slope analyses which were computed at three levels of the moderator (i.e., violence exposure): one standard deviation below the mean (low), the mean (moderate), and one standard deviation above the mean (high; Montoya & Hayes, 2017).

AFNI software was used to complete group-level analyses of fMRI data. Functional maps representing the blood oxygenation-level-dependent fMRI signal during stress versus control conditions of the MIST were compared to violence exposure, behavioral measures of stress, and mental health questionnaires using linear mixed-effect models (3dLME). Functional neuroimaging data were collected on two different MRI scanners (i.e., 3T Siemens Allegra and 3T Siemens Prisma Scanners). Thus, scanner type was included as a covariate to account for potential variance related to differences between the scanners. Further, participant sex was included as a covariate to account for variance related to sex differences in neural reactivity to stress (Cohen et al., 2023; Seo et al., 2017; Wang et al., 2007).

Neuroimaging analyses were restricted using a whole-brain gray matter mask to reduce the number of voxel-wise comparisons. The gray matter mask was created from the average anatomical images from participants in the present study (3dMean in AFNI) and segmented into gray matter, white matter, and cerebrospinal fluid (3dSeg in AFNI). White matter and cerebrospinal fluid were then excluded from the mask. The segmented gray matter mask was then used to limit subsequent fMRI analyses to the gray matter of the brain. A family-wise error (FWE) corrected, voxel-wise threshold of $p < .05$ was used to reduce Type I error in all neuroimaging analyses. The corrected cluster significance threshold was determined by Monte Carlo simulations (3dClustSim [version compiled July 9, 2016]) using an uncorrected significance threshold of $p < .005$. The smoothness of fMRI data was derived from residual volumes from the first-level analysis then averaged across subjects based on spherical autocorrelation function parameters (3dFWHMx). The results of this

simulation yielded a critical cluster extent volume threshold of 467 mm³. Additional Monte Carlo simulations were carried out for the hippocampus and amygdala using a small volume correction based on a priori hypotheses for these regions (hippocampus: 174 mm³ and amygdala: 101 mm³). Results of the Monte Carlo simulations corresponding to AFNI clusterize options for nearest neighbor one and two-sided criteria were used to determine cluster volume thresholds for each simulation. Whole-brain results are reported in Table 4. The interpretation and discussion of the fMRI results focus on significant regions of activation with a priori hypotheses (frontoparietal–hippocampal–amygdala network).

Follow-Up Analyses

Follow-up analyses were conducted upon data from regions that showed a significant three-way interaction in the 3dLME analysis. Specifically, follow-up analyses were conducted for functional regions of interest identified by the 3dLME that fell within the frontoparietal–hippocampal–amygdala network, given a priori hypotheses for these brain regions. Data were extracted (AFNI: 3dROIstats) from significant areas of activation for the control and stress conditions of the MIST. The differential (stress–control MIST) neural response from each functional regions of interest was then included in moderation analyses, using the PROCESS macro in Statistical Package for the Social Sciences (Hayes, 2012; Hayes & Preacher, 2013). Corrections for multiple comparisons were completed for the moderation analyses using Benjamini–Hochberg correction ($p_{\text{false discovery rate (FDR)}} < .05$) across the significant areas of activation for which we had a priori hypotheses (i.e., frontoparietal–hippocampal–amygdala network). PROCESS was then used to compute simple slopes for each significant moderation effect that passed Benjamini–Hochberg correction. Each simple slopes analysis examined the relationship between stress-elicited neural activity and behavioral measures (i.e., anxiety, depression, posttraumatic stress, stress rating, SCR, heart rate, and cortisol) to stress at different levels of the moderator (i.e., violence exposure): one standard deviation below the mean (low), the mean (moderate), and one standard deviation above the mean (high; Hayes, 2012).

Transparency and Openness Promotion Guidelines

Participants in this study did not consent to the release of their data to a third-party repository. Therefore, we are unable to publicly archive data due to the conditions of our ethics approval. Readers seeking access to the current data should contact the corresponding author (David C. Knight). Data can only be released to individuals who complete a formal collaboration agreement with the principal investigators. Access can only be granted in accordance with ethical procedures governing the reuse of sensitive data. No part of the study procedures or analyses

Table 2
Emotional Response to Control Versus Stress Montreal Imaging Stress Task

Measure	Stress ($M \pm SEM$)	Control ($M \pm SEM$)	<i>n</i>	<i>t</i>	<i>p</i>
Stress rating	25.78 $\pm$ 0.40	15.20 $\pm$ 0.34	288	22.10	<.001
Heart rate	73.25 $\pm$ 0.99	68.18 $\pm$ 0.81	174	9.61	<.001
Skin conductance	0.81 $\pm$ 0.05	0.38 $\pm$ 0.03	231	10.76	<.001

Note. Heart rate was measured in beats per minute (bpm). Skin conductance reflects the response to math events during the Montreal Imaging Stress Task and is presented as β values. SEM = standard error of the mean.

Table 3
Pearson Correlations of Emotional Response and Violence Exposure

Measure	1	2	3	4	5	6	7	8	9
1. Violence exposure	—								
2. Stress rating	-.26**	—							
3. Heart rate	-.04	.30**	—						
4. Skin conductance	-.05	.19**	.17	—					
5. Cortisol	-.02	.07	-.08	-.15*	—				
6. Anxiety	.21**	.07	-.02	.02	.00	—			
7. Depression	.11*	.08	-.02	-.07	.00	.53**	—		
8. Posttraumatic stress	.34**	-.09	-.07	.00	.02	.50**	.38**	—	
9. MRI scan age	.06	-.17**	-.26**	-.11	.02	-.08	-.04	.01	—

Note. MRI = magnetic resonance imaging.

* $p < .05$. ** $p < .01$.

were preregistered in a time-stamped institutional registry before the research was conducted. We have shared stimulus presentation codes, audio feedback files, and data analysis codes under <https://osf.io/dw2pj/> (Eickstead & Knight, 2024). We reported how we determined our sample size, data exclusions, all inclusion and exclusion criteria, all manipulations, and all measures in the study. Inclusion and exclusion criteria were established prior to data analysis.

Results

Stress Rating

Stress ratings were not collected from the first 13 participants. Thus, stress rating data for these participants were not included in analyses of these data. Paired t -test comparisons of stress ratings indicated that participants found the stress (25.78 ± 0.40 ; $M \pm$ standard error of

the mean) condition more stressful than the control (15.20 ± 0.34) condition, $t(287) = 22.10$, $p < .001$; Table 2.

Heart Rate

Heart rate data were not analyzed for 127 participants due to equipment malfunction or excessive noise in the signal. Thus, heart rate data were assessed for 174 participants. Paired t -test comparisons indicate that heart rate was significantly higher during the stress (73.25 ± 0.99) than control (68.18 ± 0.81) condition, $t(173) = 9.61$, $p < .001$; Table 2. These data provide psychophysiological evidence that the stress condition was more stressful than the control condition. Pearson correlations were used to assess the relationship between heart rate and stress ratings. Heart rate was positively correlated stress ratings ($r = 0.30$, $p < .01$; Table 3).

Table 4
Three-Way Interaction Between Violence Exposure, Behavioral Data, and Stress-Elicited Brain Activity

VE $\times$ Behavior $\times$ Brain	Region	Hemisphere	(x, y, z)	Volume (mm ³)	F	p
Depression	Dorsolateral PFC	Left	-32, 37, 15	1,343	15.67	.0001
	Dorsolateral PFC	Right	25, 10, 57	987	14.77	.0001
	Precuneus	Left	-8, -54, 67	509	12.62	.0004
	Inferior temporal gyrus	Left	-56, -37, -25	516	20.72	<.00001
	Cerebellum	Left	-42, -67, -48	546	13.56	.0003
	Cerebellum	Left	-25, -68, -22	500	17.49	.00004
Anxiety	Cerebellum	Left	-7, -50, -46	503	12.70	.0004
Posttraumatic stress	Superior occipital gyrus	Right	26, -78, 29	582	14.82	.0001
Skin conductance	Dorsolateral PFC	Left	-20, 8, 64	723	17.88	.00003
	Dorsolateral PFC	Left	-25, 52, 30	501	16.27	.00008
	Dorsomedial PFC	Left	-2, 13, 62	572	16.06	.00008
	Ventrolateral PFC	Right	32, 30, -15	903	19.23	.00002
	Paracentral lobule	Right	8, -33, 74	1,759	21.46	<.00001
	Cuneus	Right	9, -92, 22	892	13.98	.0002
	Fusiform face complex	Right	43, -55, -20	1,858	16.72	.00006
	Inferior temporal gyrus	Left	-48, -45, 18	1,451	13.30	.0003
	Primary visual cortex	Right	18, -95, -8	531	15.05	.0001
	Inferior occipital gyrus	Left	-38, -86, -6	1,236	19.64	.00002
	Ventral vision complex	Left	-25, -49, -18	533	14.89	.0001
	Cerebellum	Right	8, -81, -17	748	12.49	.0005
	Dorsomedial PFC	Left	-1, 44, 33	3,912	15.36	.0001
	Dorsolateral PFC	Left	-23, 51, 16	1,187	16.34	.0001
	Middle temporal gyrus	Right	50, 1, -22	562	14.28	.0002

Note. Location, F -statistic, and MNI coordinates of the peak voxel for significant areas of activation. Cluster thresholds for analyses were set to whole brain (467 mm^3), hippocampus (174 mm^3), and amygdala (101 mm^3) at $p = .005$. VE = violence exposure; PFC = prefrontal cortex; MNI = Montreal Neurologic Institute.

Skin Conductance Response

Six participants did not have SCR data due to equipment failure, and 64 participants had no measurable SCRs (i.e., zero SCRs above $0.05 \mu\text{Siemens}$). Therefore, SCR data were assessed for 231 participants. Paired *t*-test comparisons revealed that SCRs to math events were significantly greater during the stress (0.81 ± 0.05) than control (0.38 ± 0.03) condition of the MIST, $t(230) = 10.76$, $p < .001$; Table 2. These data provide additional psychophysiological evidence that the stress condition was more stressful than the control condition. Pearson correlations demonstrated that SCR was positively correlated with stress ratings ($r = 0.19$, $p < .01$; Table 3). SCR did not vary with heart rate.

Cortisol

Paired *t*-test comparisons of salivary cortisol revealed that baseline cortisol (0.22 ± 0.007) was significantly greater than post-MIST cortisol (0.19 ± 0.007), $t(298) = 6.30$, $p < .001$; Table 2. This is an unexpected finding that may be attributable to environmental, temporal, and social factors that have been observed and discussed in prior research (Allendorfer et al., 2014; Gossett et al., 2018; Orem et al., 2019). Cortisol levels varied negatively with SCR ($r = -0.15$, $p < .05$). However, cortisol levels did not vary with stress ratings or heart rate (Table 3).

Violence Exposure

Violence exposure was assessed at Waves 1–4 ($n = 301$; 0.85 ± 0.04 , $M \pm SEM$; Table 1). Violence exposure data were positively skewed (skewness = 0.935, kurtosis = 0.424, Kolmogorov–Smirnov $Z = 0.108$, $p < .001$). However, the violence exposure data were assumed to approximate normality due to a sufficiently large sample size ($N = 301$), according to the central limit theorem (Islam, 2018; Kwak & Kim, 2017; Nimon, 2012). Pearson correlations assessed the relationship between violence exposure and other measures assessed in this study. Violence exposure was negatively correlated with stress ratings ($r = -0.26$, $p < .01$). Specifically, the greater the violence exposure, the lower the differential (stress–control MIST) stress ratings. Violence exposure was not correlated with stress-elicited heart rate, SCR, or cortisol (Table 3).

Square-root and logarithmic transformations were completed individually on the violence exposure variable, given that violence exposure data were not normally distributed. The logarithmic transformation slightly decreased the skewness (skewness = 0.714) and kurtosis (kurtosis = 0.071) values. Further, the square-root transformation decreased the skewness value (skewness = 0.189) and increased the kurtosis value (kurtosis = 0.619). However, the violence exposure data were still statistically nonnormally distributed (i.e., $p < .001$ using the Shapiro–Wilk test of normality), regardless of transformation type. Pearson correlations assessed the relationship between the transformed (i.e., square-root and logarithmic) violence exposure data and the other variables of interest. Similar to the original analysis, the square-root-transformed ($r = -0.25$, $p < .01$) and logarithm-transformed ($r = -0.18$, $p < .01$) data were both negatively correlated with stress ratings. However, neither transformed variable (i.e., square-root and logarithmic) was correlated with stress-elicited heart rate, SCR, or cortisol. Thus, the interpretation of results remained unchanged, and the original violence exposure variable was retained in all subsequent analyses,

given the results were not affected by the square-root or logarithmic transformations.

Anxiety

Anxiety symptoms were assessed at Wave 4 ($n = 301$; 2.71 ± 2.06). Pearson correlations assessed the relationship between anxiety symptoms and other variables collected for this study. Anxiety symptoms were positively correlated with violence exposure ($r = 0.21$, $p < .01$). Specifically, the greater the violence exposure, the greater the symptoms of anxiety. Anxiety did not vary with stress rating, heart rate, SCR, or cortisol (Table 3).

Depression

Depression symptoms were assessed at Wave 4 ($n = 301$; 1.93 ± 1.59). Pearson correlations evaluated the relationship between symptoms of depression and other measures assessed in the present study. Depression symptoms varied positively with violence exposure ($r = 0.11$, $p < .05$) and symptoms of anxiety ($r = 0.53$, $p < .01$; Table 2). Depression did not vary with stress ratings, heart rate, SCR, or cortisol (Table 3).

Posttraumatic Stress

Posttraumatic stress data were assessed at Wave 4. Data were not collected for nine participants. Therefore, analyses were restricted to participants with complete data ($n = 292$; 0.46 ± 0.54). Pearson correlations evaluated the relationship between posttraumatic stress symptoms and other measures collected during this study. Posttraumatic stress symptoms varied positively with violence exposure ($r = 0.34$, $p < .01$), symptoms of anxiety ($r = 0.50$, $p < .01$), and symptoms of depression ($r = 0.38$, $p < .01$). Posttraumatic stress symptoms did not vary with stress ratings, heart rate, SCR, or cortisol (Table 3).

Violence Exposure and Stress-Elicited Behavior

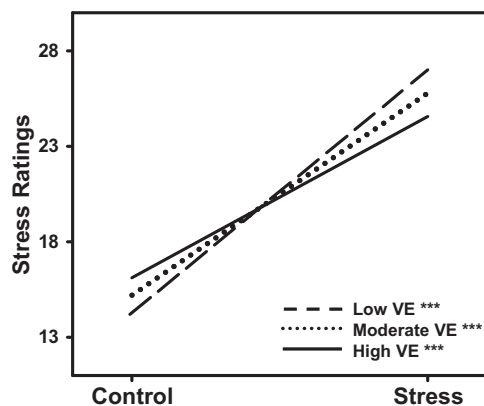
Violence Exposure and Stress Ratings

Moderation analysis assessed whether violence exposure moderated the relationship between MIST condition and stress ratings. The analysis revealed a significant effect of violence exposure on differential stress ratings (i.e., stress–control MIST; $\beta = -3.21$, $p < .001$). Specifically, the greater the exposure to violence, the lower the differential stress ratings. Thus, violence exposure moderated the relationship between MIST condition and stress ratings. The simple slope analysis revealed that the relationship between MIST condition (i.e., stress vs. control) and stress ratings was stronger in participants with low violence exposure ($\beta = 12.73$, $p < .001$) than participants with moderate ($\beta = 10.59$, $p < .001$) and high levels of violence exposure ($\beta = 8.45$, $p < .001$), although higher stress ratings were observed in the stress than control condition across all levels of the moderator (Figure 1).

Violence Exposure and Heart Rate

The moderation analysis revealed no effect of violence exposure on differential heart rate ($\beta = 0.49$, $p > .05$). Thus, the relationship between MIST condition and heart rate did not vary with the amount of violence exposure.

Figure 1
MIST Stress Ratings Moderated by Violence Exposure



Note. The graph depicts the relationship between MIST condition (i.e., stress vs. control) and stress ratings, which were moderated by violence exposure. Dashed (low), dotted (moderate), and solid (high) lines represent the amount of violence exposure. MIST = Montreal Imaging Stress Task; VE = violence exposure.

*** $p < .001$.

Violence Exposure and SCR

The moderation analysis revealed no effect of violence exposure on differential SCR ($\beta = 0.04$, $p > .05$). Thus, the relationship between MIST condition and SCR did not vary with the amount of violence exposure.

Violence Exposure and Cortisol

The moderation analysis revealed no effect of violence exposure on differential cortisol levels ($\beta = 0.01$, $p > .05$). Thus, the relationship between MIST condition and cortisol levels did not vary with the amount of violence exposure.

Functional MRI

Linear mixed-effect models (AFNI: 3dLME) were used to assess the relationship between the differential fMRI signal to the stress versus control conditions of the MIST, violence exposure, and behavioral data (i.e., anxiety, depression, posttraumatic stress, stress ratings, SCR, heart rate, and cortisol levels). The results of the whole-brain 3dLME analysis have been reported in Table 4. The interpretation and discussion of the 3dLME results below focus on significant areas of activation (i.e., from the 3dLME) that fall within the frontoparietal–hippocampal–amygdala network, based on our *a priori* hypotheses. Moderation analyses (PROCESS) were used to assess the relationship between behavioral data and the neural response to stress at three levels of the moderator, violence exposure (i.e., low: 1 *SD* below the mean, moderate: the mean, and high: 1 *SD* above the mean).

Violence Exposure, Stress-Elicited Brain Activity, and Stress Ratings

Linear mixed-effect models assessing the interaction between MIST condition, violence exposure, and stress ratings revealed no

significant areas of activation within the frontoparietal–hippocampal–amygdala network ($p > .05$). Therefore, moderation analyses were not completed.

Violence Exposure, Stress-Elicited Brain Activity, and Heart Rate

Linear mixed-effect models assessing the interaction between MIST condition, violence exposure, and heart rate revealed no significant areas of activation in the frontoparietal–hippocampal–amygdala network ($p > .05$). Therefore, moderation analyses were not completed.

Violence Exposure, Stress-Elicited Brain Activity, and SCR

Linear mixed-effect models revealed a significant interaction between MIST Condition (i.e., stress vs. control MIST), violence exposure, and SCR. Specifically, stress-elicited right ventrolateral PFC activity varied with violence exposure and SCR, $F(1, 227) = 19.23$, $p^{\text{FWE}} < .05$; Table 4. Follow-up moderation analyses revealed that prior violence exposure moderated the relationship between stress-elicited brain activity and SCR, $F(1, 227) = 19.40$, $p^{\text{FDR}} < .05$; Table 5. Specifically, stress-elicited right ventrolateral PFC activity varied positively with SCR in participants with low ($\beta = 0.07$, $p < .0001$) and moderate levels of violence exposure ($\beta = 0.02$, $p = .0237$). In contrast, those with high levels of violence exposure showed no relationship between SCR and right ventrolateral PFC activity ($p > .05$; Figure 2; Table 5).

Linear mixed-effect models also revealed that stress-elicited activity within the dorsomedial PFC varied with violence exposure and SCR, $F(1, 227) = 16.06$, $p^{\text{FWE}} < .05$; Table 4. Follow-up moderation analyses demonstrated that prior violence exposure moderated the relationship between stress-elicited dorsomedial PFC activity and SCR, $F(1, 227) = 10.96$, $p^{\text{FDR}} < .05$; Table 5. Specifically, dorsomedial PFC activity varied positively with SCR in participants with low levels of violence exposure ($\beta = 0.04$, $p = .0085$), while dorsomedial PFC activity varied negatively with SCR in participants with high levels of violence exposure ($\beta = -0.03$, $p = .0407$). No relationship was observed between dorsomedial PFC activity and SCR in participants with moderate levels of violence exposure ($p > .05$; Figure 2; Table 5).

Similar to the dorsomedial PFC, stress-elicited activity within two regions of the left dorsolateral PFC varied with violence exposure and SCR, $F(1, 227) = 17.88$, $p^{\text{FWE}} < .05$; $F(1, 227) = 16.27$, $p^{\text{FWE}} < .05$; Table 4. Follow-up moderation analysis revealed that prior violence exposure moderated the relationship between stress-elicited left dorsolateral PFC activity and SCR, $F(1, 227) = 11.85$, $p^{\text{FDR}} < .05$; $F(1, 227) = 13.51$, $p^{\text{FDR}} < .05$; Table 5. Specifically, activity in one region of the left dorsolateral PFC varied positively with SCR in participants with low levels violence exposure ($\beta = 0.05$, $p = .0006$), while there was no relationship in participants with moderate or high violence exposure ($p > .05$). Similarly, activity in the second region of the dorsolateral PFC varied positively with SCR in participants with low levels of violence exposure ($\beta = 0.03$, $p = .0109$). In contrast, activity within the second region of the dorsolateral PFC varied negatively with SCR in participants with high levels of violence exposure ($\beta = -0.03$, $p = .0078$). No relationship was observed between left dorsolateral PFC activity and SCR in participants with moderate levels of violence exposure ($p > .05$; Figure 2; Table 5).

Table 5
Brain and Behavior Relationships Moderated by Violence Exposure

Behavior	Region	Level of violence exposure	β	df	t	p
SCR	Dorsolateral PFC*	Low	0.05	228	3.48	.0006
		Moderate	0.01		1.57	.1178
		High	−0.02		−1.39	.1658
	Dorsolateral PFC*	Low	0.03	228	2.57	.0109
		Moderate	0.00		0.00	1.0000
		High	−0.03		−2.69	.0078
	Dorsomedial PFC*	Low	0.04	228	2.66	.0085
		Moderate	0.01		0.50	.6157
		High	−0.03		−2.06	.0407
	Ventrolateral PFC*	Low	0.07	228	4.64	.0000
		Moderate	0.02		2.28	.0237
		High	−0.02		−1.59	.1138
Cortisol	Dorsolateral PFC*	Low	0.19	296	3.00	.0030
		Moderate	−0.03		−0.61	.5454
		High	−0.25		−3.14	.0019
	Dorsomedial PFC*	Low	0.17	296	2.52	.0122
		Moderate	−0.04		−0.73	.4658
		High	−0.24		−2.90	.0040
Depression	Dorsolateral PFC*	Low	−0.01	300	−2.32	.0212
		Moderate	0.00		0.84	.4030
		High	0.01		3.70	.0003
	Dorsolateral PFC*	Low	−0.01	300	−2.61	.0096
		Moderate	−0.00		−0.36	.7164
		High	0.01		2.26	.0246

Note. Moderation analyses were restricted to significant areas of activation from the 3dLME analysis (Table 4) for which we had a priori hypotheses (i.e., dorsolateral, dorsomedial, and ventrolateral PFC). β represents the slope of the relationship between behavior and stress-elicited neural activity at low, moderate, and high levels of violence exposure. SCR = skin conductance response; PFC = prefrontal cortex; 3dLME = linear mixed-effect models. All moderation results pass Benjamini–Hochberg corrections (* $p_{\text{false discovery rate}} < .05$).

Violence Exposure, Stress-Elicited Brain Activity, and Cortisol

Linear mixed-effect models revealed that stress-elicited dorsomedial PFC, $F(1, 295) = 15.36, p^{\text{FWE}} < .05$, and dorsolateral PFC, $F(1, 295) = 16.34, p^{\text{FWE}} < .05$, activity varied with violence exposure and cortisol levels (Table 4). Moderation analyses revealed that prior violence exposure moderated the relationship between stress-elicited activity within these brain regions and cortisol, dorsomedial: $F(1, 295) = 12.55, p^{\text{FDR}} < .05$; dorsolateral: $F(1, 295) = 15.93, p^{\text{FDR}} < .05$; Table 5. Specifically, both dorsomedial ($\beta = 0.17, p = .0122$) and dorsolateral ($\beta = 0.19, p = .003$) PFC activity varied positively with cortisol levels in participants with low levels of violence exposure, while dorsomedial ($\beta = -0.24, p = .004$) and dorsolateral ($\beta = -0.25, p = .0019$) PFC activity varied negatively with cortisol levels in participants with high violence exposure. There was no relationship between cortisol and activity within these regions in participants with moderate levels of violence exposure ($p > .05$; Figure 3; Table 5).

Violence Exposure, Stress-Elicited Brain Activity, and Anxiety

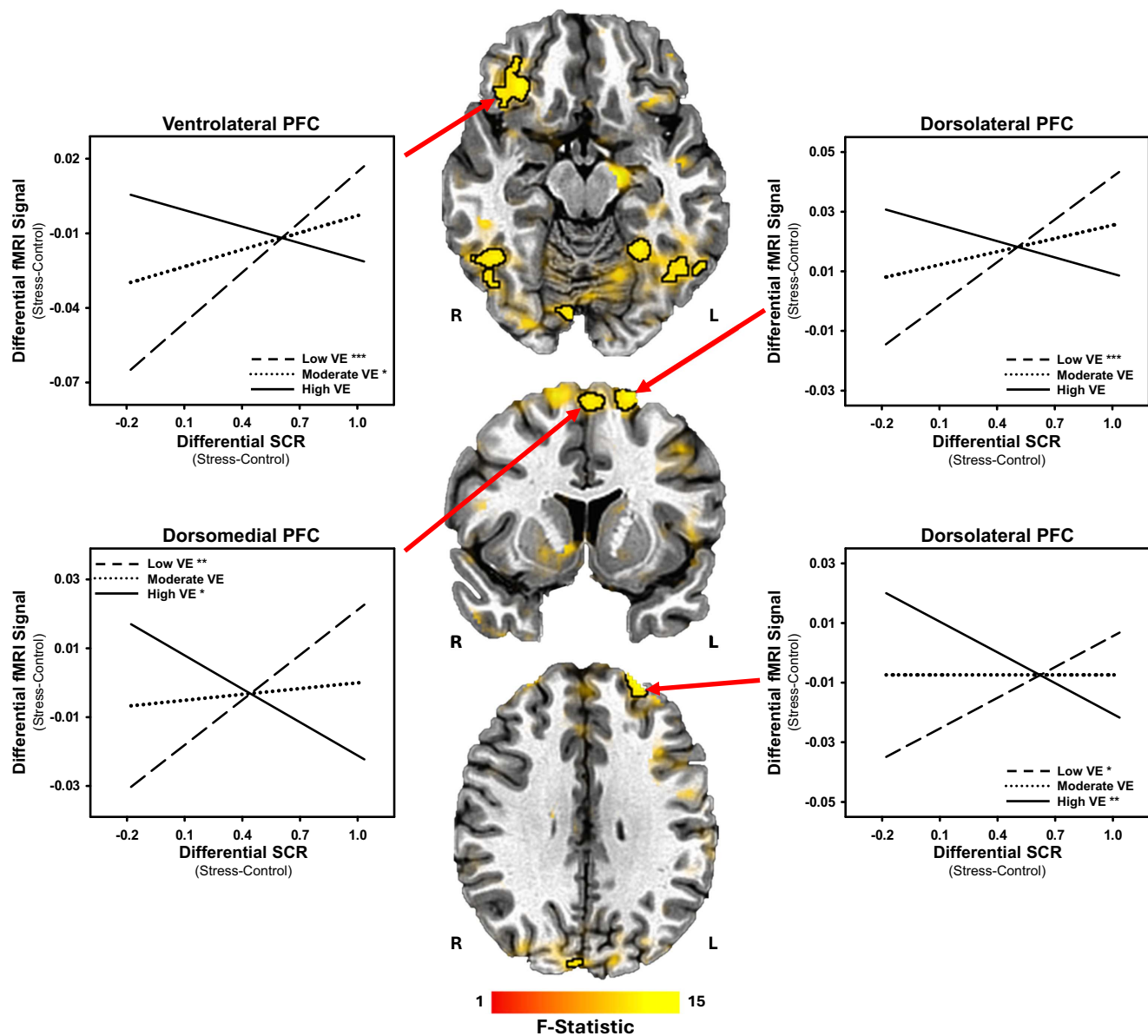
Linear mixed-effect models assessing the interaction between MIST condition, violence exposure, and anxiety symptoms revealed no significant areas of activation within the frontoparietal–hippocampal–amygdala network ($p > .05$). Therefore, moderation analyses were not completed.

Violence Exposure, Stress-Elicited Brain Activity, and Depression

Linear mixed-effect analysis revealed that stress-elicited (i.e., stress vs. control) activity within the bilateral dorsolateral PFC varied with violence exposure and depression symptoms, left: $F(1, 297) = 15.67, p^{\text{FWE}} < .05$, and right: $F(1, 297) = 14.77, p^{\text{FWE}} < .05$; Table 4. Moderation analyses demonstrated that prior violence exposure moderated the relationship between stress-elicited dorsolateral PFC activity and depression, left: $F(1, 297) = 17.52, p^{\text{FDR}} < .05$, and right: $F(1, 297) = 11.70, p^{\text{FDR}} < .05$; Table 5. Specifically, dorsolateral PFC activity varied negatively with depression in participants with low levels of violence exposure (left: $\beta = -0.01, p = .0212$, and right: $\beta = -0.01, p = .0096$). In contrast, dorsolateral PFC activity varied positively with depression in participants with high levels of violence exposure (left: $\beta = 0.01, p = .0003$, and right: $\beta = 0.01, p = .0246$). There was no relationship between bilateral dorsolateral PFC activity and depression in participants with moderate levels of violence exposure ($p > .05$; Figure 4; Table 5).

Violence Exposure, Stress-Elicited Brain Activity, and Posttraumatic Stress

Linear mixed-effect models assessing the interaction between MIST condition, violence exposure, and posttraumatic stress revealed no significant areas of activation in the frontoparietal–hippocampal–amygdala network ($p > .05$). Therefore, moderation analyses were not completed.

Figure 2*Brain Activity and Skin Conductance Response Relationship Moderated by Violence Exposure*

Note. Brain images depict the interaction between MIST condition (i.e., stress vs. control), violence exposure, and SCR (AFNI: 3dLME). Results are presented at a family-wise error corrected threshold of $p < .05$. Suprathreshold areas of activation are fully opaque and outlined with a black border. Subthreshold areas of activation are visible with increasing opacity as statistical significance increases. The line graphs depict the results of the moderation analysis, which assessed whether the relationship between brain activity (stress–control MIST) and SCR (stress–control MIST) was moderated by violence exposure. Dashed (low), dotted (moderate), and solid (high) lines represent the amount of violence. MIST = Montreal Imaging Stress Task; SCR = skin conductance response; AFNI = Analysis of Functional NeuroImages; 3dLME = linear mixed-effect models; VE = violence exposure; PFC = prefrontal cortex; fMRI = functional MRI. See the online article for the color version of this figure.

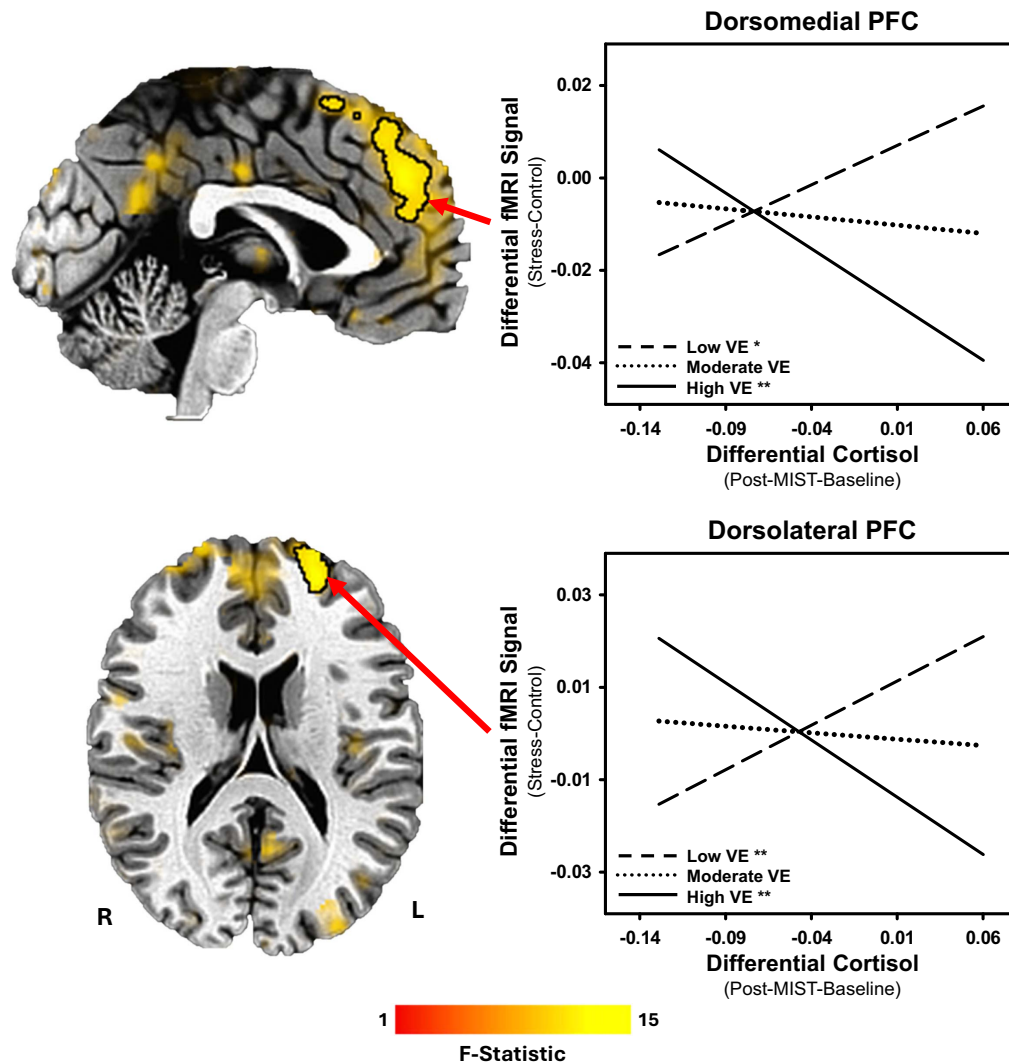
* $p < .05$. ** $p < .01$. *** $p < .001$.

Discussion

Adolescent violence exposure has adverse effects on emotional health that often persist into adulthood (Buckholdt et al., 2015; Cook et al., 2018; Heinze et al., 2017; Heleniak et al., 2018; Pattwell & Bath, 2017). Determining the relationship between violence exposure and emotion-related neurobehavioral function could have

important implications for our understanding of psychopathology. Therefore, the present study investigated the interrelationships between prior violence exposure, emotional behavior (i.e., anxiety, depression, posttraumatic stress, stress ratings, SCR, heart rate, and cortisol), and stress-elicited neural activity (i.e., fMRI). We found that higher levels of violence exposure were associated with greater symptoms of depression, anxiety, and posttraumatic stress. We also found that the

Figure 3
Brain Activity and Cortisol Relationship Moderated by Violence Exposure

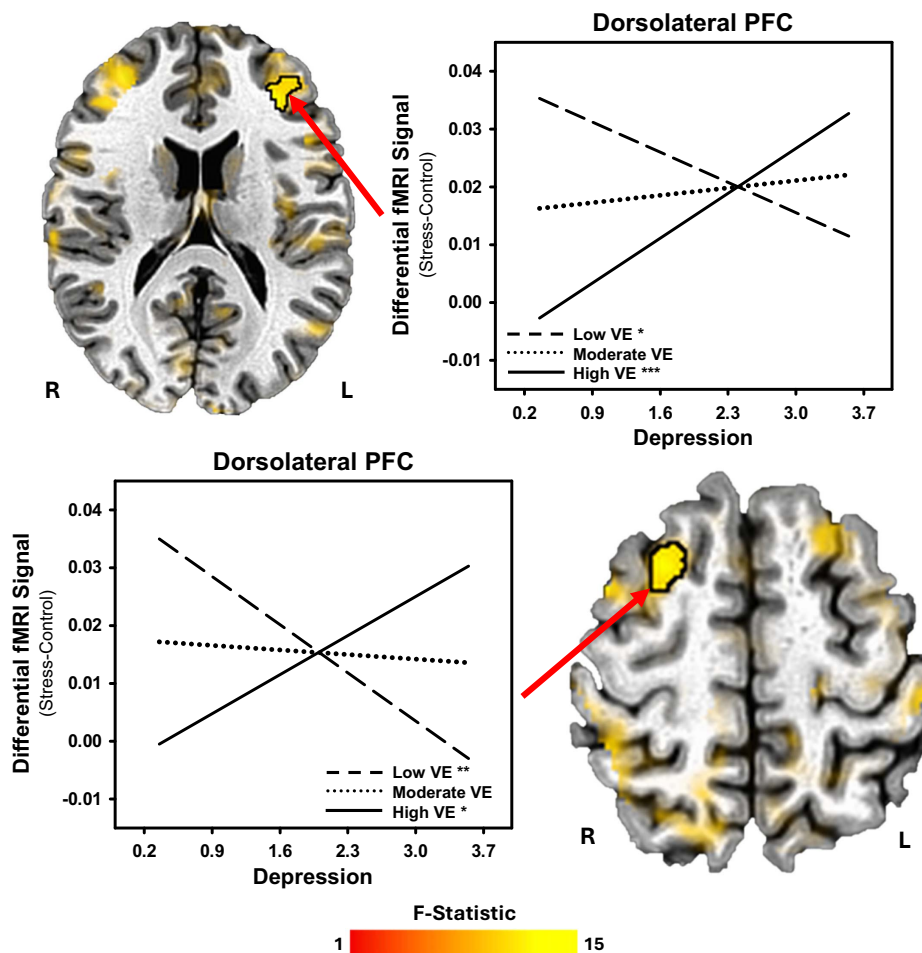


Note. Brain images depict the interaction between MIST condition (i.e., stress vs. control), violence exposure, and cortisol (AFNI: 3dLME). Results are presented at a family-wise error corrected threshold of $p < .05$. Suprathreshold areas of activation are fully opaque and outlined with a black border. Subthreshold areas of activation are visible with increasing opacity as statistical significance increases. The line graphs depict the results of the moderation analysis, which assessed whether the relationship between brain activity (stress–control MIST) and cortisol (Post-MIST–Baseline) was moderated by violence exposure. Dashed (low), dotted (moderate), and solid (high) lines represent the amount of violence exposure. MIST = Montreal Imaging Stress Task; AFNI = Analysis of Functional NeuroImages; 3dLME = linear mixed-effect models; VE = violence exposure; PFC = prefrontal cortex; fMRI = functional MRI. See the online article for the color version of this figure.

* $p < .05$. ** $p < .01$.

amount of prior violence exposure moderated the relationship between stress-elicited neural activity and emotion function. Specifically, we found that violence exposure moderated the relationship between dorsolateral PFC activity and mental health (i.e., depressive symptoms). Additionally, violence exposure moderated the relationship between dorsolateral, dorsomedial, and ventrolateral PFC activity and psychophysiology (i.e., SCR and cortisol). These findings suggest that exposure to violence alters the relationship between neural function and emotional behavior (i.e., mental health and psychophysiology).

The current findings suggest that violence exposure may alter the relationship between stress-related neural activity and emotional function. Specifically, violence exposure moderated the relationship between depression and stress-elicited dorsolateral PFC activity. Greater depression was associated with greater stress-elicited dorsolateral PFC activity in participants with high levels of violence exposure. In contrast, greater depression was associated with lower stress-elicited dorsolateral PFC activity in participants with low levels of violence exposure. Thus, these findings suggest that the

Figure 4*Brain Activity and Depression Relationship Moderated by Violence Exposure*

Note. Brain images depict the interaction between MIST condition (i.e., stress vs. control), violence exposure, and depression (AFNI: 3dLME). Results are presented at a family-wise error corrected threshold of $p < .05$. Suprathreshold areas of activation are fully opaque and outlined with a black border. Subthreshold areas of activation are visible with increasing opacity as statistical significance increases. The line graphs depict the results of the moderation analysis, which assessed whether the relationship between brain activity (stress-control MIST) and depression was moderated by violence exposure. Dashed (low), dotted (moderate), and solid (high) lines represent the amount of violence exposure. MIST = Montreal Imaging Stress Task; AFNI = Analysis of Functional NeuroImages; 3dLME = linear mixed-effect models; VE = violence exposure; PFC = prefrontal cortex; fMRI = functional MRI. See the online article for the color version of this figure.

* $p < .05$. ** $p < .01$. *** $p < .001$.

relationship between neural function and depression varies with the amount of prior violence exposure. Specifically, participants with greater prior violence exposure showed greater stress-elicited dorsolateral PFC activity when they reported more depression symptoms. On the other hand, participants with less prior violence exposure showed less stress-elicited dorsolateral PFC activity when they reported more depression symptoms. Previous research has also observed relationships between depression and emotion-related dorsolateral PFC activity. Specifically, adult participants with depression showed less dorsolateral PFC activity during negative (i.e., aversive pictures and words) than neutral and positive stimuli (Davidson, 2000; Siegle et al., 2002). The current findings, however, expand our understanding of the relationship between dorsolateral PFC activity

and depression. In the present study, we found that the relationship between stress-related dorsolateral PFC activity and depression was moderated by the amount of violence exposure. The current findings, in combination with prior research, suggest that the relationship stress-elicited dorsolateral PFC activity has with depressive symptoms varies with the amount of prior violence exposure.

Violence exposure also moderated the relationship between stress-elicited dorsolateral, dorsomedial, and ventrolateral PFC activity and SCR. SCR varied positively with PFC (i.e., dorsolateral, dorsomedial, and ventrolateral) activity in participants with low levels of violence exposure. Further, left dorsolateral and dorsomedial PFC activity varied negatively with SCR in participants with high levels of violence exposure. These findings suggest

that greater amounts of violence exposure may modify healthy neurobehavioral relationships. Specifically, these findings suggest that PFC activity may underlie healthy regulation of peripheral emotion (i.e., SCR), and this mechanism may be moderated by prior exposure to violence. Previous research suggests that the PFC regulates the peripheral expression of the emotional response (Allaert et al., 2022; Critchley et al., 2000; Herrmann et al., 2018; Orem et al., 2019; Patterson et al., 2002). For example, greater activity in the dorsolateral, dorsomedial, and ventrolateral PFC is associated with greater SCR (Critchley et al., 2000; Orem et al., 2019; Patterson et al., 2002). Further, low-amplitude transcranial direct current stimulation of the dorsolateral and ventrolateral PFC elicits lower amplitude SCRs than sham stimulation (Allaert et al., 2022; Herrmann et al., 2018). The current findings also suggest that SCR varies with PFC activity. However, our findings further our understanding of these relationships by demonstrating that the relationship between PFC activity and SCR is moderated by violence exposure. Prior research on adverse life experiences is generally consistent with the current findings (Harnett et al., 2019; Purcell et al., 2021; Teicher & Samson, 2016). For example, prior work suggests that adverse childhood experiences (e.g., violence exposure and sexual abuse) alter PFC function and physiological responses to threat or stress (Harnett et al., 2018, 2019; Purcell et al., 2021). Therefore, the present findings, in combination with prior work, suggest that the relationship between stress-related PFC (i.e., dorsolateral, dorsomedial, and ventrolateral) activity and the peripheral emotional response to stress (e.g., SCR) is moderated by the level of prior violence exposure.

Violence exposure also moderated the relationship between stress-elicited dorsolateral and dorsomedial PFC activity and cortisol. Stress-elicited dorsolateral and dorsomedial PFC activity varied positively with cortisol among those with low levels of violence exposure. In contrast, stress-elicited dorsolateral and dorsomedial PFC activity varied negatively with cortisol in participants with high violence exposure. These findings suggest that greater violence exposure may modify the neuroendocrine response to stress. More specifically, while the dorsolateral and dorsomedial PFC may support the regulation of stress hormones (e.g., cortisol), violence exposure appears to moderate the relationship between PFC activity and stress hormone release. Several prior studies have also observed a relationship between dorsolateral and dorsomedial PFC function and the peripheral emotional response (e.g., SCR, heart rate, cortisol; Kern et al., 2008; Keulers et al., 2015; Peters et al., 2016; Wang et al., 2005; Wheelock et al., 2016). For example, prior research has observed a negative relationship between dorsomedial PFC activity and stress-elicited cortisol release (Kern et al., 2008; Keulers et al., 2015). Other research has found a positive relationship between dorsolateral PFC activity and stress-elicited cortisol release (Kern et al., 2008; Peters et al., 2016; Wang et al., 2005). This prior research suggests there is a relationship between cortisol and dorsolateral and dorsomedial PFC activity. The current findings, in combination with this prior work, suggest the dorsomedial and dorsolateral PFC may support emotion processes that ultimately regulate the peripheral expression of emotion (e.g., cortisol). Although generally consistent with prior research, conclusions about the current findings should be tempered with the understanding that the group-level cortisol results did not show a stress-related increase (i.e., baseline vs. post-MIST) in the present study. In fact, cortisol decreased from baseline to post-MIST. As noted above, the decrease in cortisol from baseline to post-MIST was an unexpected finding that may be attributable to

environmental, temporal, and social factors that have been discussed in prior work (Allendorfer et al., 2014; Gossett et al., 2018; Orem et al., 2019). Further, the cortisol response to the MIST did not vary with violence exposure. Still, our findings do suggest that violence exposure moderated the relationship between dorsomedial and dorsolateral PFC activity and cortisol levels. These findings suggest the relationship between dorsolateral and dorsomedial PFC function and the peripheral emotional response (i.e., cortisol) may be moderated by prior exposure to violence.

Violence exposure among young people may significantly alter the development and function of the brain. In fact, the present findings suggest that violence exposure moderates the relationship between brain activity and psychophysiology (e.g., SCR and cortisol). For example, we observed a relatively typical relationship between PFC activity and psychophysiological responses (i.e., SCR and cortisol) among those exposed to low levels of violence. Specifically, the greater the brain activity, the greater the psychophysiological response. These findings are broadly consistent with other research that has found a positive relationship between PFC function and psychophysiology (Allaert et al., 2022; Critchley et al., 2000; Herrmann et al., 2018; Kern et al., 2008; Peters et al., 2016; Wang et al., 2005). Taken together with this prior work, the current findings suggest that those exposed to low levels of violence show a relatively typical and healthy relationship between PFC function and psychophysiological activity. However, the relationship between PFC function and psychophysiology appeared to shift among young people exposed to higher levels of violence (i.e., moderate and high levels in the present study). More specifically, no relationship was generally observed between PFC activity and psychophysiology at moderate levels of violence exposure. This finding suggests the PFC may play a rather limited role in the expression and regulation of the peripheral emotional response in those exposed to moderate levels of violence, which is consistent with the view that violence exposure desensitizes young people to the effects of future stressors (Gaylord-Harden et al., 2011; Kennedy & Ceballo, 2016; McCart et al., 2007). The relationship between PFC activity and psychophysiology may shift even further among those exposed to the highest levels of violence. Specifically, we found a somewhat mixed relationship between PFC activity and psychophysiology among those exposed to high levels of violence. While several regions of the PFC showed no relationship with psychophysiology, others showed a negative relationship among those exposed to high levels of violence. Repeated exposure to high levels of violence may shift psychological processes (e.g., vigilance, threat appraisal, rumination, and desensitization) supported by the PFC and may interact, in a complex manner, with other neural networks (e.g., frontoparietal, salience, and limbic networks) that support the peripheral expression of emotion (Arnsten, 2009; Dark et al., 2020, 2023; Heleniak et al., 2018; Kennedy & Ceballo, 2016; Kliever & Lepore, 2015; Kliever & Sullivan, 2008; Woo et al., 2021). For example, regulatory control over psychophysiological activity may gradually shift to other brain regions (e.g., amygdala) among those exposed to higher levels of violence (Arnsten, 2009; Woo et al., 2021). Thus, the peripheral emotional response may show a blunted or negative relationship with PFC function among young people exposed to the highest levels of violence. The present findings suggest that violence exposure moderates the relationship between brain activity and psychophysiology, perhaps by altering the brain function and connectivity that supports psychological processes. These changes may ultimately modify neural connectivity and the

function of other brain regions, altering emotional expression and regulation.

We also found that behavioral responses (i.e., differential stress ratings) were moderated by violence exposure. Stress ratings were higher in the stress than control condition of the MIST. However, the relationship between the MIST and stress ratings was stronger among participants with low levels of violence exposure than participants with moderate or high levels of violence exposure (Figure 1). In contrast, stress-elicited physiological responses (i.e., SCR, heart rate, and cortisol) were not moderated by violence exposure. Physiological responses (i.e., SCR, heart rate, and cortisol) are often considered an automatic, reflexive expression of stress, while stress ratings represent the cognitive perception of stress. These findings suggest that, while participants with greater levels of violence exposure may cognitively perceive the MIST as less stressful than participants with lower levels of violence exposure, physiological responses are not moderated in the same manner.

The present study suggests that those exposed to more violence experience more internalizing symptoms (i.e., depression, anxiety, and posttraumatic stress). Prior research is generally consistent with our findings (Buckholdt et al., 2015; Cook et al., 2018; Heinze et al., 2017; Kundu et al., 2018; Lo Iacono & Carola, 2018; Pattwell & Bath, 2017). These prior studies suggest that the positive relationship observed between violence exposure and internalizing symptoms in the present study may be due to the disruption of emotion regulation processes. Specifically, violence exposure may disrupt allostasis, the process through which an organism responds to changes in the environment by altering normal, healthy regulatory function (Danese & McEwen, 2012; Ramsay & Woods, 2014). The disruption of allostatic systems found in prior work may underlie depression, anxiety, and posttraumatic stress in participants with greater violence exposure. Thus, the relationship between allostatic processes and mental health symptoms may be moderated by prior violence exposure.

Limitations

The results of the present study should be considered in light of a few limitations. Specifically, the present study assessed violence exposure across four waves, but fMRI data were only collected once (i.e., after Wave 4). Longitudinal neuroimaging research may provide additional insight into the interrelationship between violence exposure, stress-elicited neural activity, and emotional function. Additionally, the present study did not include socioeconomic status and substance use (i.e., alcohol, tobacco, and cannabis) as covariates in fMRI analyses. Higher levels of violence exposure are often found in neighborhoods with lower socioeconomic status and greater disadvantage (Chong et al., 2015; De Coster et al., 2006; Gibson et al., 2009; Kravitz-Wirtz et al., 2022). Further, higher levels of violence exposure have been linked with greater substance use (Beharie et al., 2019; Löfving-Gupta et al., 2018; Vermeiren et al., 2003). Therefore, future studies may benefit from accounting for variance in socioeconomic status and substance use. Furthermore, mental health questionnaires used in the present study (e.g., DISC and RCMAS) were administered across all waves of the Healthy Passages study to provide continuity across each timepoint. However, these mental health questionnaires were designed for children and adolescents and have not been validated in adult (i.e., age 20–23 years in the present study) samples. Future studies into the interrelationship between violence exposure, brain function, and emotion may benefit from addressing these limitations.

Conclusions

In conclusion, violence exposure appears to moderate important aspects of the relationship between PFC activity and the emotional response to stress. More specifically, the present results suggest the relationship between dorsolateral, dorsomedial, and ventrolateral PFC activity and emotional function varies with the amount of prior violence exposure. The current findings, in combination with prior work, suggest that these brain regions play an important role in the expression and regulation of the emotional response to stress and underlie individual variability in stress reactivity (Buckholdt et al., 2015; Casey et al., 2019; Harnett et al., 2019; Orem et al., 2019; Silvers & Guassi Moreira, 2019; Teffer & Semendeferi, 2012; Wheelock et al., 2016). Thus, stress-elicited neural reactivity and emotion-related processes may be moderated by violence exposure. The present findings link violence exposure to emotion-related behaviors (i.e., depression, SCR, and cortisol), supporting the view that violence exposure alters neural and emotional function. These findings provide novel evidence that violence exposure modulates the relationship between emotion-related behaviors and stress-elicited brain activity.

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